American Computer Science League

2025-2026 ● Contest 3: Short Problems Solutions ● Senior Division

1. Boolean Algebra

$A\,(\overline{B} + C) + \overline{B}(\overline{A} + C) + C(A + B)$
$= A\overline{B} + AC + \overline{B} * \overline{A} + \overline{B}C + CA + CB$
$= A\overline{B} + AC + \overline{B} * \overline{A} + \overline{B}C + BC$
$= \overline{B}(A + \overline{A} + C) + AC + BC$
$= \overline{B} + AC + BC$
All terms must be FALSE. Therefore $B = 1$ and $C = 0$ and $A = *$.
Two ordered triples (*,1,0) make it FALSE.

B. 2

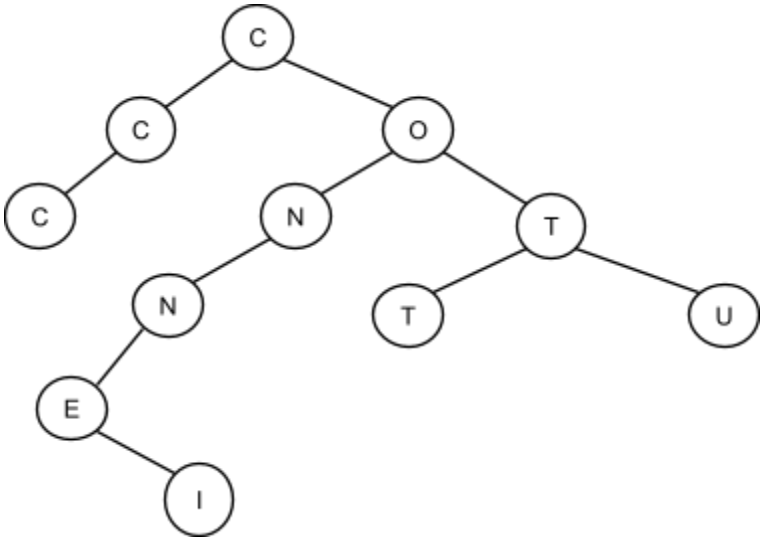
2. Boolean Algebra

$\overline{ABC} + \overline{\overline{B} * \overline{A + C}} + \overline{C}(A + B)$
$= \overline{A} + \overline{B} + \overline{C} + B + A + C + A\overline{C} + B\overline{C}$
Using the first 6 terms, it is always TRUE so the other 2 don't matter.

E. 1

3. Data Structures

The binary search tree for **CONNECTICUT** is:



The breadth-first traversal of this tree is CCOCNTNTUEI.

A. CCOCNTNTUEI

4. Data Structures This algorithm performed on a stack pushes each letter onto the stack unless it is a double letter. Therefore, ORANGE is the only string without a double letter.	D. ORANGE
5. FSAs It must begin and end with an a. It is possible for the string to not include a 'b' by using the top branch, a 'c'using the middle branch, or a 'd' using the bottom branch. Remember that a loop means 0 or more times.	A. a
6. Regular Expressions The license plates that match **`[D-R1-5]* [^0-2][a-t]?`** are J44ee - valid EN811f - stops at 1 IR54p - valid FO795 - stops at 7 PH94n - stops at 4 GM223g - valid Dan - valid 48ACSL - stops at A That's 4 or half of them.	C. half of them